

Paulo V. Galvao Simplicio

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EDUCATION

- **West Virginia University** Morgantown, WV
Ph.D. in Robotics Engineering — GPA: 4.00/4.00 Jan. 2023 – Dec. 2026
- **University of São Paulo** São Paulo, Brazil
M.Sc. in Electrical Engineering (Dynamical Systems) — GPA: 4.00/4.00 Aug. 2019 – Dec. 2021
- **Tiradentes University** Maceió, Brazil
B.Eng. in Mechatronics Engineering — GPA: 8.98/10.0, Top of Class Jan. 2014 – Dec. 2018

TECHNICAL SKILLS

- **Languages:** Python, C++, MATLAB
- **Robotics & Simulation:** ROS2, Gazebo, NVIDIA Isaac Sim / Isaac Lab, RViz
- **Motion Planning:** A*, RRT*, OMPL, octree / voxel-grid planners, multi-resolution replanning, behavior trees
- **Perception & State Estimation:** Open3D, OpenCV, point-cloud processing, GTSAM (factor graphs), GNSS/IMU sensor fusion, visual odometry
- **Control:** PID, LQR, MPC, disturbance-observer-based robust control
- **Machine Learning:** Reinforcement Learning (PPO), PyTorch, neural networks
- **Tools & Platforms:** Linux, Git, SolidWorks, Raspberry Pi, Arduino
- **Certifications:** FAA Remote Pilot (Part 107)
- **Spoken Languages:** English, Portuguese

RESEARCH EXPERIENCE

- **West Virginia University — Robotics Lab** Morgantown, WV
Graduate Research Assistant, Ph.D. Candidate (advisor: G. A. S. Pereira) Jan. 2023 – Present
 - Lead UAV autonomy research on a DOE-funded program for autonomous inspection of coal-ash and tailings dams, owning the planning, perception-aware mapping, and onboard replanning stack.
 - Designed a multi-resolution motion planner (A*/RRT* on octrees, OMPL, Open3D) that generates collision-free “tunnel” corridors for UAV inspection in cluttered environments; published at *ICUAS 2025*.
 - Built a photogrammetry-aware coverage planner enforcing GSD, image-overlap, row-spacing, and view-geometry constraints to produce flight plans for high-fidelity 3D reconstruction of large structures; published at *ICUAS 2024*.
 - Engineered real-time online replanning with hazard detection and return-to-home (RTH) logic, enabling battery-swap missions over large sites without operator intervention.
 - Training PPO policies in NVIDIA Isaac Lab for autonomous next-best-view (NBV) planning, optimizing UAV trajectories on raw point clouds under photogrammetric constraints to maximize new coverage (*ongoing*).
 - Developing real-time state estimation by fusing visual odometry with GNSS/IMU in factor graphs (GTSAM) for consistent pose tracking and map alignment in GPS-degraded conditions (*ongoing*).
- **University of São Paulo — Center for Robotics (CROB)** São Paulo, Brazil
Graduate Researcher, M.Sc. Electrical Engineering Aug. 2019 – Dec. 2021
 - Developed robust and intelligent quadrotor controllers (PID, LQR, MPC, and disturbance-observer-based) for trajectory tracking under wind gusts and parametric uncertainty; validated in Gazebo and on hardware.
 - Results published in *IET Control Theory & Applications* (2024) and *IEEE Access* (2022).

SELECTED PROJECTS & COMPETITIONS

- **XPRIZE Rainforest — Brazilian Team (Finalist)** International
May 2023 – 2024
Robotics Subteam, University of São Paulo
 - Co-developed a ruggedized remotely-operated mobile robot for autonomous biodiversity sampling (leaf litter, soil) in tropical rainforests.
 - Deployed and tested platforms in field campaigns in Brazil and Singapore, collaborating with biologists from Brazil and France on species identification and biodiversity analysis.
- **SARC–BARINet Aerospace Competition — Flying U2 Team (International Winner)** International
Aug. 2019 – Dec. 2021
University of São Paulo
 - Designed and tuned cascaded UAV controllers (PID, LQR, MPC) for autonomous flight over water under high-wind disturbance.
 - Validated the full stack in Gazebo and in field trials, contributing to the team’s international win and a peer-reviewed publication.

SELECTED PUBLICATIONS

Full list on Google Scholar.

- **P. V. G. Simplicio**, J. R. S. Benevides, R. S. Inoue, M. H. Terra. “Robust and Intelligent Control of Quadrotors Subject to Wind Gusts.” *IET Control Theory & Applications* 18(18):2594–2611, 2024.
- K. C. T. Vivaldini, T. F. Pazelli, L. G. S. Rocha, I. A. D. Santos, K. A. Q. Caldas, D. P. Soler, **P. V. G. Simplicio**, *et al.* “Multi-UAV Collaborative System for the Identification of Surface Cyanobacterial Blooms and Aquatic Macrophytes.” *Journal of Intelligent & Robotic Systems* 110(1):40, 2024.
- J. R. S. Benevides, M. A. D. Paiva, **P. V. G. Simplicio**, R. S. Inoue, M. H. Terra. “Disturbance Observer-Based Robust Control of a Quadrotor Subject to Parametric Uncertainties and Wind Disturbance.” *IEEE Access* 10:7554–7565, 2022.
- **P. V. G. Simplicio**, G. A. S. Pereira. “Multi-Resolution UAV Path Replanning for Inspection of Tailings Dams.” *ICUAS*, Charlotte, NC, pp. 256–263, 2025.
- **P. V. G. Simplicio**, G. A. S. Pereira. “Mission Planning for Photogrammetry-Based Autonomous 3D Mapping of Dams Using a Commercial UAV.” *ICUAS*, Chania, Greece, pp. 464–471, 2024.
- B. M. Rocamora, **P. V. G. Simplicio**, G. A. S. Pereira. “A Behavior Tree Approach for Battery-Aware Inspection of Large Structures Using Drones.” *ICUAS*, Chania, Greece, pp. 234–240, 2024.
- **P. V. G. Simplicio**, J. R. S. Benevides, M. A. D. Paiva, R. S. Inoue, M. H. Terra. “Disturbance Observer-Based Control for Trajectory Tracking of a Quadrotor Subject to Uncertainties.” *Congresso Brasileiro de Automática (CBA)*, 2022.

TEACHING & INDUSTRY EXPERIENCE

- **University of São Paulo** São Paulo, Brazil
Jul. 2022 – Dec. 2022
Teaching Assistant — Graduate-Level Artificial Intelligence
 - Led lab sections and office hours, graded projects, and refined lab materials and rubrics with the instructor.
- **Sennze Automation** Brazil
May 2015 – Dec. 2015
Hardware & Embedded Software Intern
 - Contributed to an automated wireline retrieval system with encoder-based depth tracking and safety alarms to prevent pull-head shoot-outs.

AWARDS & HONORS

- **Jack and Marietta Mullenger Fellowship** — West Virginia University, 2024–2025.
- **SARC–BARINet Aerospace Competition — International Winner**, 2021.
- **XPRIZE Rainforest — Finalist** (Brazilian Team), 2024.
- **Graduated Top of Class**, B.Eng. Mechatronics — Tiradentes University, 2018.